Machine Learning Concepts 1 (CSE3967)

Assignment-2

Hands-On Exercises

Serial Number	Questions	Course Outcome
1	Using the dataset, *Student_Performance_Prediction .csv*, perform the following: (a) Load the dataset and report the column names, number of samples, number of features, and target variable. Visualise the relationship between any two numerical features using a scatter plot. (b) Calculate the Pearson correlation coefficient between every pair of numerical features and display the resulting correlation matrix. (c) Calculate the Pearson correlation coefficient between each numerical feature and the target variable and report the corresponding correlation coefficients. (d) Based on the result in 1(c), rank the features according to the absolute strength of their correlation with the target, and identify the most influential feature based on correlation. (e) Visualise the relationship between the target variable and the top three features identified in (d) using appropriate scatter plots and interpret the observed relationships.	CO1, CO2

2	Perform feature selection based on correlation analysis using the results obtained in Question 1 (a) Based on the correlation matrix obtained in 1(b), identify highly correlated features that may contain redundant information using a suitable correlation threshold, e.g., $\|r\| \geq 0.70$. (b) Based on the result in 1(c), determine which features have a strong correlation with the target using $\|r\| \geq 0.40$. (c) For each pair of highly correlated features identified in 2(a), retain the feature having the stronger absolute correlation with the target and remove the other feature as redundant. (d) Create a modified dataset by retaining the features selected in 2(b) after removing any redundant features identified in 2(a) and 2(c). Save the resulting dataset as *modified_Student_Performance.csv*.	CO2
3	Using the train-test split method, apply Linear Regression to both the original dataset and the modified dataset created in 2(d) to predict the target variable (`Final_Score`). (a) Calculate and compare suitable regression error metrics, including MAE, MSE, RMSE, and R^2, for the original and modified datasets. (b) Print the regression equations for both datasets. (c) Calculate the prediction error for the test data as: Prediction Error = Actual Value−Predicted Value Plot the prediction errors against the test-sample index using a line plot for both the original and modified datasets. Analyze the plots and explain whether the prediction errors are small or large and whether the linear regression models provide a good fit. (d) Plot the actual and predicted *Final_Score* values for the test data using a line plot for both the original and modified datasets. Analyse both the plots.	CO1,CO2

4	**Using the original dataset and the train-test split method, perform the following tasks:** (a) Apply Polynomial Regression to the dataset using polynomial degrees 2, 3, and 4 separately. (b) Calculate and report the training MSE, testing MSE, training R^2, and testing R^2 for each polynomial degree. (c) Plot the training MSE and testing MSE against the polynomial degree (2, 3, and 4) using a line plot and analyze the results for possible underfitting or overfitting. (d) Plot the actual versus predicted *Final_Score* values using scatter plots for each polynomial degree (2, 3, and 4), and identify the degree that provides the best overall fit.	CO1 ,CO2, CO3

Home Assignments

Serial Number	Questions	Course Outcome
1	**Using the *modified_Student_Performance.csv* dataset, perform the following:** (a) Create a new feature named *Academic_Activity* by adding *Practice_Tests* and *Assignments_Completed*: $$Academic_Activity = Practice_Tests + Assignments_Completed.$$ (b) Calculate the Pearson correlation coefficient of *Academic_Activity* with *Final_Score* and compare it with the Pearson correlation coefficients of *Practice_Tests* and *Assignments_Completed* with *Final_Score*. Identify which of the three features has the strongest linear association with *Final_Score*. (c) Add the new feature to the dataset and save the resulting dataset as *feature_engineered_Student_Performance.csv*.	CO1, CO2

2	(a) Split the new dataset created above into training and testing sets and train Polynomial Regression models with polynomial degrees 2 and 3 separately. (b) Calculate the training and testing MSE and R^2 for both polynomial models. (c) Prepare a comparison table containing the training and testing MSE and R^2 of the two polynomial models. (d) Identify the model with the lowest training MSE, the lowest testing MSE, and the highest testing R^2. (e) Determine whether any of the models show signs of overfitting or underfitting. Justify your answer by comparing their training and testing performance. (f) Suppose the model will be used to predict the *Final_Score* of new students. Which model would you choose for deployment? Justify your choice based on testing performance and generalization. (g) Using the selected polynomial model, predict the *Final_Score* for three new students and display the predicted scores in a table. (h) Does increasing the polynomial degree necessarily improve prediction performance on unseen data? Explain your answer based on the results obtained for the two polynomial models.	CO2, C03